

DQL Queries for Kubernetes

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Advanced Query Patterns for Kubernetes Data

This notebook provides a comprehensive reference of DQL queries for Kubernetes monitoring. From basic entity queries to complex performance analysis, these patterns help you extract insights from your Kubernetes data.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Kubernetes monitoring
Permissions	<code>metrics.read</code> , <code>entities.read</code> , <code>logs.read</code>
Knowledge	K8S-01 through K8S-07
Data	Active Kubernetes cluster monitored

1. Entity Queries

Kubernetes Entity Types

Entity Type	DQL Table	Description
Cluster	<code>dt.entity.kubernetes_cluster</code>	K8s clusters
Node	<code>dt.entity.kubernetes_node</code>	Worker nodes
Namespace	<code>dt.entity.cloud_application_namespace</code>	Namespaces
Workload	<code>dt.entity.cloud_application</code>	Deployments, StatefulSets
Container	<code>dt.entity.container_group_instance</code>	Container instances
Service	<code>dt.entity.service</code>	Detected services

```
// List all Kubernetes clusters
fetch dt.entity.kubernetes_cluster
| fields entity.name, tags
| sort entity.name asc
```

```
// List all Kubernetes nodes
fetch dt.entity.kubernetes_node
| fields entity.name, tags
| sort entity.name asc
```

```
// List all namespaces
fetch dt.entity.cloud_application_namespace
| fields entity.name, tags
| sort entity.name asc
```

```
// List all workloads (deployments, statefulsets)
fetch dt.entity.cloud_application
| fields entity.name, tags
| sort entity.name asc
| limit 50
```

```
// Count Kubernetes nodes
fetch dt.entity.kubernetes_node
| summarize nodeCount = count()
```

2. Metric Queries

Common Kubernetes Metrics

Metric	Description	Unit
<code>dt.containers.cpu.usage_percent</code>	Container CPU vs limit	Percent
<code>dt.containers.memory.usage_percent</code>	Container memory vs limit	Percent
<code>dt.containers.cpu.throttled_time</code>	CPU throttle time	Nanoseconds
<code>dt.containers.memory.working_set_bytes</code>	Working memory	Bytes
<code>dt.kubernetes.workload.requests_cpu</code>	CPU requests	Millicores
<code>dt.kubernetes.workload.requests_memory</code>	Memory requests	Bytes

```
// Container CPU usage – top consumers
fetch dt.metrics
| filter metric.key == "dt.containers.cpu.usage_percent"
| summarize avgCpu = avg(value), by:{dt.entity.container_group_instance}
| sort avgCpu desc
| limit 15
```

```
// Container memory usage approaching limits
fetch dt.metrics
| filter metric.key == "dt.containers.memory.usage_percent"
| summarize avgMem = avg(value), by:{dt.entity.container_group_instance}
| filter avgMem > 80
| sort avgMem desc
```

```
// CPU throttling detection
fetch dt.metrics
| filter metric.key == "dt.containers.cpu.throttled_time"
| summarize totalThrottle = sum(value), by:
{dt.entity.container_group_instance}
| filter totalThrottle > 0
| sort totalThrottle desc
| limit 15
```

```
// Namespace-level resource usage
fetch dt.metrics
| filter metric.key == "dt.containers.cpu.usage_percent"
| summarize avgCpuUsage = avg(value), by:{k8s.namespace.name}
| sort avgCpuUsage desc
| limit 15
```

```
// Resource requests by namespace (capacity planning)
fetch dt.metrics
| filter metric.key == "dt.kubernetes.workload.requests_cpu"
| summarize avgCpuRequests = avg(value), by:{k8s.namespace.name}
| sort avgCpuRequests desc
| limit 15
```

3. Log and Event Queries

Log Query Patterns

```
// Error logs by namespace
fetch logs
| filter loglevel == "ERROR"
| filter isNotNull(k8s.namespace.name)
| summarize errorCount = count(), by:{k8s.namespace.name}
| sort errorCount desc
| limit 15
```

```
// Kubernetes events – warnings only
fetch logs
| filter matchesPhrase(content, "Warning")
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source, "k8s")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

```
// Pod crashes and restarts
fetch logs
| filter matchesPhrase(content, "CrashLoopBackOff") or matchesPhrase(content, "BackOff")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// OOM events
fetch logs
| filter matchesPhrase(content, "OOMKilled") or matchesPhrase(content, "Out of memory")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// Log volume trend by namespace
fetch logs, from: now() - 24h
| filter isNotNull(k8s.namespace.name)
| summarize log_count = count(), by:{k8s.namespace.name, time_bucket =
bin(timestamp, 1h)}
| sort time_bucket asc
| limit 200
```

4. Trace Queries

Span Queries for Kubernetes Services

```
// Service response times in K8s
fetch spans
| filter span.kind == "server"
| filter isNotNull(k8s.namespace.name)
| summarize
    p50 = percentile(duration, 50),
    p95 = percentile(duration, 95),
    p99 = percentile(duration, 99),
    by:{k8s.namespace.name, dt.entity.service}
| sort p99 desc
| limit 15
```

```
// Error rate by service
fetch spans
| filter span.kind == "server"
| filter isNotNull(k8s.namespace.name)
| summarize
    total = count(),
    errors = countIf(otel.status_code == "ERROR"),
    by:{k8s.namespace.name, dt.entity.service}
| fieldsAdd errorRate = 100.0 * toDouble(errors) / toDouble(total)
| filter errors > 0
| sort errorRate desc
| limit 15
```

```
// Slow traces (>1 second)
fetch spans
| filter span.kind == "server" and duration > 1000000000
| filter isNotNull(k8s.namespace.name)
| fields timestamp, trace.id, k8s.namespace.name, span.name, duration
| sort duration desc
| limit 20
```

```
// Request throughput by namespace
fetch spans
| filter span.kind == "server"
| filter isNotNull(k8s.namespace.name)
| summarize requestCount = count(), by:{k8s.namespace.name}
| sort requestCount desc
| limit 10
```

5. Correlation Queries

Joining Multiple Data Sources

```
// High CPU workloads with their names
fetch dt.metrics
| filter metric.key == "dt.containers.cpu.usage_percent"
| summarize avgCpu = avg(value), by:{dt.entity.cloud_application}
| filter avgCpu > 50
| lookup [fetch dt.entity.cloud_application | fields id, entity.name],
sourceField:dt.entity.cloud_application, lookupField:id
| fields entity.name, avgCpu
| sort avgCpu desc
```

```
// Nodes with high utilization
fetch dt.metrics
| filter metric.key == "dt.kubernetes.node.cpu_usage"
| summarize avgNodeCpu = avg(value), by:{dt.entity.kubernetes_node}
| filter avgNodeCpu > 70
| lookup [fetch dt.entity.kubernetes_node | fields id, entity.name],
sourceField:dt.entity.kubernetes_node, lookupField:id
| fields entity.name, avgNodeCpu
| sort avgNodeCpu desc
```

6. Dashboard Queries

Queries Optimized for Dashboards

```
// Cluster summary tile
fetch dt.entity.kubernetes_cluster
| summarize clusterCount = count()
```

```
// Node count tile
fetch dt.entity.kubernetes_node
| summarize nodeCount = count()
```

```
// Error rate trend (chart)
fetch logs, from: now() - 24h
| filter loglevel == "ERROR" and isNotNull(k8s.namespace.name)
| summarize errors = count(), by:{time_bucket = bin(timestamp, 1h)}
| sort time_bucket asc
```

```
// Top namespaces by CPU (bar chart)
fetch dt.metrics
| filter metric.key == "dt.containers.cpu.usage_percent"
| summarize avgCpu = avg(value), by:{k8s.namespace.name}
| sort avgCpu desc
| limit 10
```

7. Alerting Queries

Threshold-Based Alerting Patterns

```
// High memory usage alert query
fetch dt.metrics
| filter metric.key == "dt.containers.memory.usage_percent"
| summarize avgMem = avg(value), by:{dt.entity.container_group_instance}
| filter avgMem > 90
| summarize alertCount = count()
```



```
// CrashLoop detection (last hour)
fetch logs, from: now() - 1h
| filter matchesPhrase(content, "CrashLoopBackOff")
| summarize crashCount = count()
```

```
// Error rate spike detection
fetch logs, from: now() - 1h
| filter loglevel == "ERROR" and isNotNull(k8s.namespace.name)
| summarize errors = count(), by:{k8s.namespace.name}
| filter errors > 100
| sort errors desc
```

8. Query Templates

Reusable Query Patterns

Filter by Namespace:

```
| filter k8s.namespace.name == "NAMESPACE_NAME"
```

Filter by Cluster:

```
| filter kubernetesClusterName == "CLUSTER_NAME"
```

Time-based Analysis:

```
fetch logs, from: now() - DURATION
| summarize log_count = count(), by:{time_bucket = bin(timestamp, INTERVAL)}
| sort time_bucket asc
```

Top N Pattern:

```
| sort METRIC desc
| limit N
```

Summary

This notebook provided DQL query patterns for:

- Entity discovery and relationships
 - Container and node metrics
 - Log and event analysis
 - Distributed tracing
 - Multi-source correlation
 - Dashboard visualizations
 - Alerting conditions
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References

- [DQL Documentation](#)
 - [Kubernetes Metrics Reference](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.